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Paleomagnetism of the Late Permian–Early Triassic Koipato Volcanics, Nevada: implications for latitudinal displacement

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We have obtained paleomagnetic directions for 11 basalt flows of the probable Early Triassic Koipato Volcanics from the South Tobin Range of western Nevada. The mean direction for these flows ($D = 350.3°$; $I = 39.0°$; $\alpha_{95} = 8.0°$) is discordant with respect to the APW path for North America. A positive fold test, a consistency-of-reversal test, the shape and amount of between-site dispersion, and other considerations strongly suggest that the magnetization is pre-folding and probably primary. The most likely explanation for this discordance is southward transport and clockwise rotation of the sampling area. The most probable amount of displacement and net rotation are 1400 km and 25°, respectively. These observations have considerable significance for the early Mesozoic tectonics of the Great Basin.

1. Introduction

The Mesozoic tectonics of western Nevada has been a subject of long debate, and continues to have considerable significance in discussions of the evolution of the North American continental margin and the histories of accreted terranes [1–6]. Much discussion centers on the nature and timing of the early Mesozoic Sonoma orogeny. We have obtained paleomagnetic results that provide some useful constraints.

The Sonoma orogeny is the oldest recognized tectonic event of probable Mesozoic age in western Nevada (Fig. 1). The Sonoma orogeny is dated by deformed rocks as young as Guadelupian within the Golconda allochthon, and by the existence of an unconformity at the base of Triassic strata that cover the deformed Golconda rocks. The basal unit of the cover sequence is the Koipato volcanics [3,7], consisting of mafic and felsic extrusive rocks. The Koipato is dated by Spathian ammonites that occur high in the section [8], and by overlying Spathian strata; it is possible that the Koipato includes older Triassic or even latest Permian rocks.

During the Sonoma orogeny the Golconda allochthon was displaced at least 80 km eastward across the edge of North America. Major questions about the emplacement of the Golconda allochthon include: (1) was it far travelled before emplacement [3], or was it locally derived [2]; (2) was it emplaced in Early Triassic or Late Permian time, before deposition of its Triassic cover [3], or was it emplaced together with its cover in post-Triassic time [5]? We reasoned that by obtaining paleolatitude data for the Koipato and other cover rocks, we might help answer these questions. To this end we have sampled and studied over 500 samples (about 60 sites) from two general areas in western Nevada. However, the difficulties of doing paleomagnetism in a multiply-deformed, much-intruded terrain are such that to date we have reliable results only from 11 sites in a single (Koipato) formation. The Koipato data are presented here; preliminary results for the remaining formations are reported in two theses [9,10]. Work on the latter rocks is still in progress.

2. Geology of the sampling area

Our results are from rocks in the southern Tobin Range (Fig. 1). In the southern Tobin

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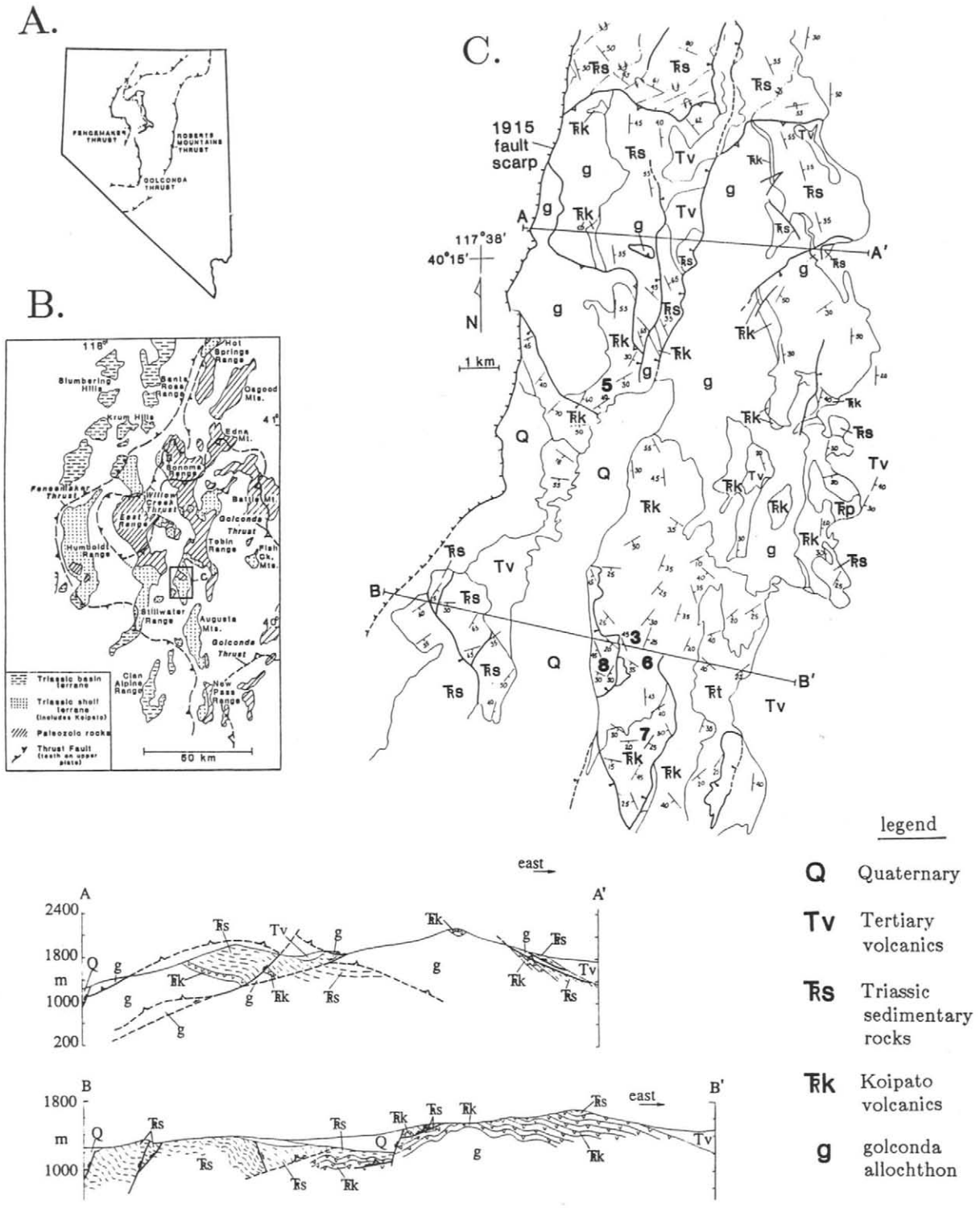


Fig. 1. A. Map of Nevada showing trace of Golconda and other regional thrusts; teeth on upper plate. B. Geologic map of north-central Nevada, showing local trace of Golconda thrust and facies of Triassic cover to Golconda allochthon; Koipato volcanics are basal unit of Triassic shelf terrane; box indicates area studied in southern Tobin Range. C. Geologic map and sections of southern Tobin Range by R. Speed, C. Bina, and P. Lundgrien (unpublished data, 1984) and Burke [19]; numbers indicate paleomagnetic sampling sites.

Range, the Koipato consists of approximately 300 m of basic and felsic tuffs, breccias and lavas that lie unconformably on the Golconda allochthon. Koipato rocks in the southern Tobin Range are undated, but are overlain conformably by the Spathian Tobin Formation; the rocks we sampled thus are Early Triassic or Late Permian (245–270 Ma [11]). The Koipato in the southern Tobin Range was involved in post-Triassic deformation; structures consist of: (1) open to closed 25° south-plunging folds; (2) local thrusts with northerly throw; (3) high and low-angle Cenozoic normal faults with westerly slip and eastward tilts of about 35°. Fortunately for our purposes, the southern Tobin Range has not experienced the widespread plutonism of much of western Nevada; only a single small body of post-Triassic intrusive rock is recognized (Fig. 1), and conodont alteration in Triassic units above the Koipato is 3–4 [12], indicating temperatures of less than about 300° C.

3. Paleomagnetic results

Results for our 11 sites are given in Table 1; all of these are basalt flows that appear fresh, massive, and internally undeformed. Mean directions and statistics under several assumptions about relative timing of magnetization, tilt, and plunge are given in Table 2.

The 11 sites are located in five general areas (Fig. 1) spaced a few tens of kilometers apart; two of the sampling areas consist of only a single flow. Several localities have flows with both magnetic polarities. Fig. 3 shows site-mean directions (reverse magnetizations inverted) before tectonic correction, and after complete tectonic correction (removal of a plunge of 25° in the direction S10° W, followed by removal of residual tilt). We regard these magnetizations as most likely primary for the following reasons:

(1) Correction for tilt decreases dispersion significantly at the 95% confidence level ($k_2/k_1 = 2.26$ [12]).

(2) The distribution is nearly circular after tilt correction; scatter size is commensurate with normal geomagnetic secular variation ($k = 33.2$; $\Delta = 13.4°$).

(3) Much smaller dispersion within-site ($k = 100$) than between-site and the presence of opposite polarities at adjacent sites (Fig. 2) argue for rapid (e.g., initial cooling) acquisition of magnetization.

(4) Normal and reverse magnetizations are antiparallel at the 95% confidence level [13]. For these reasons we believe that the tilt-corrected Koipato magnetizations give the direction of the geomagnetic field at the time the Koipato rocks formed.

If the Koipato volcanics formed in their present position relative to North America our mean di-

TABLE 1

Directions of remanent magnetization, Koipato Volcanics, southern Tobin Range, Nevada

Flow	In situ D/I (°)	Corrected for tilt				
		simple D/I (°)	complete D/I (°)	N	k	α_{95} (°)
K303	178.1/−17.9	162.8/−14.7	152.0/−14.7	10	105.9	4.7
K304	198.0/−20.8	183.5/−36.5	177.8/−36.5	5	114.3	7.2
K305	189.2/−21.2	173.2/−37.7	167.7/−37.7	4	289.0	5.4
K306	0.3/−9.8	2.2/+28.7	5.9/+28.7	6	390.9	3.4
K503	149.9/−3.6	161.3/−44.8	170.9/−44.8	4	54.1	12.6
K504	143.2/−5.2	153.7/−48.6	163.3/−48.6	4	35.5	15.6
K601	0.0/+10.8	0.0/+40.8	1.1/+40.8	6	119.9	6.1
K604	341.3/+6.0	342.9/+43.9	347.7/+43.9	7	87.1	6.5
K605	174.6/+0.9	173.3/−24.8	174.4/−24.8	8	62.0	7.1
K701	325.4/+28.1	336.8/+49.0	342.1/+49.0	12	78.5	4.9
K801	339.1/−2.6	342.0/+54.2	349.2/+54.2	14	28.2	7.6

D/I = declination/inclination (positive down); N = number of samples; k = precision parameter; α_{95} = radius of circle of 95% confidence. Tilt corrections made in two ways: (1) simple; rotate bed to horizontal about local strike; (2) complete; remove regional plunge of 190/25, then rotate residual dip to horizontal.

TABLE 2

Mean directions and paleomagnetic poles, Koipato Volcanics, southern Tobin Range, Nevada

MTC	D (°)	I (°)	α_{95} (°)	k	Latitude	Longitude
In situ	348.9	9.5	12.3	14.7	53.3° N	81.2° E
Simple	346.7	38.9	8.3	31.5	68.6° N	98.4° E
Complete	350.3	39.0	8.0	33.2	70.2° N	89.9° E

MTC = method of tilt correction (see Table 1); Latitude, Longitude = latitude and longitude of paleomagnetic pole (calculated for a site latitude of 40.1° N, 242.5° E).

rection ought to match an expected direction calculated from the apparent polar wander (APW) path for North America. This comparison is made in Fig. 4, with unmistakable results; only under various highly improbable circumstances (discussed below) can the Koipato direction from the southern Tobin Range be regarded as concordant with the APW path for North America. For any Late Permian or Early Triassic age of magnetization the mean Koipato direction is discordant in both declination and inclination. Using the 250 Ma reference pole of Gordon et al. [14]:

$R = 27.0 \pm 6.7°$; $P = -13.3 \pm 6.3°$

where R and P are the rotation and poleward transport, respectively. (Calculations done in

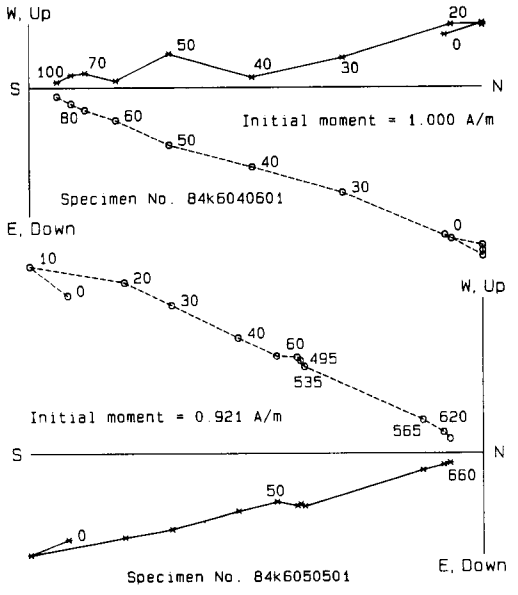


Fig. 2. Examples of demagnetization behavior of typical Koipato specimens. Orthogonal demagnetization diagrams for two adjacent, oppositely magnetized sites. Corrected for tilt, * = Map View; ○ = sections. Top: AF demagnetization from 10 to 100 mT removes single normal component that dominates magnetization of a breccia unit. Bottom: Removal of a single, reverse component by AF and thermal demagnetization is consistent with primary thermochemical magnetization during high-temperature oxidation.

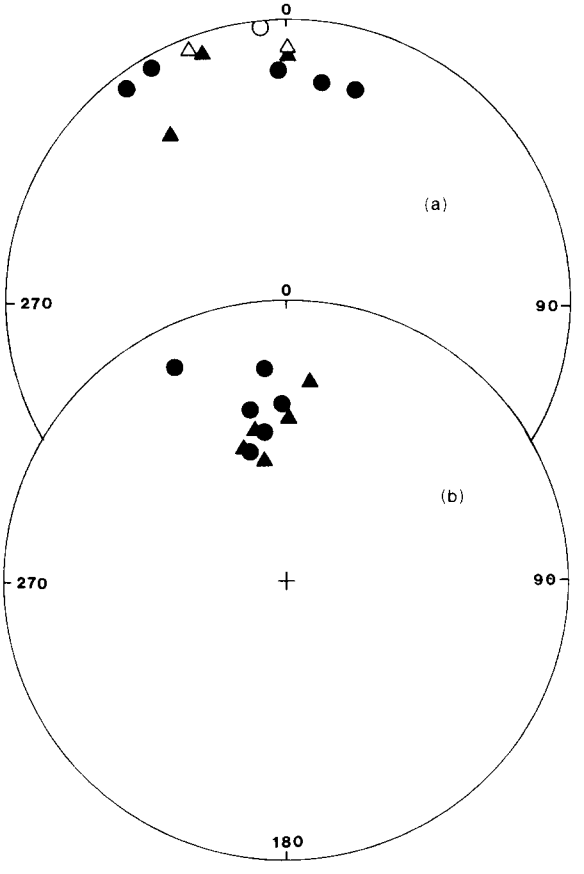


Fig. 3. Site-mean directions of remanent magnetization for Koipato basalt flows, South Tobin Range, Nevada. Triangles indicate reverse-polarity directions that have been inverted through the center of the projection. Hollow symbols, upper hemisphere. (a) In-situ directions; (b) directions after correction for tilt and plunge.

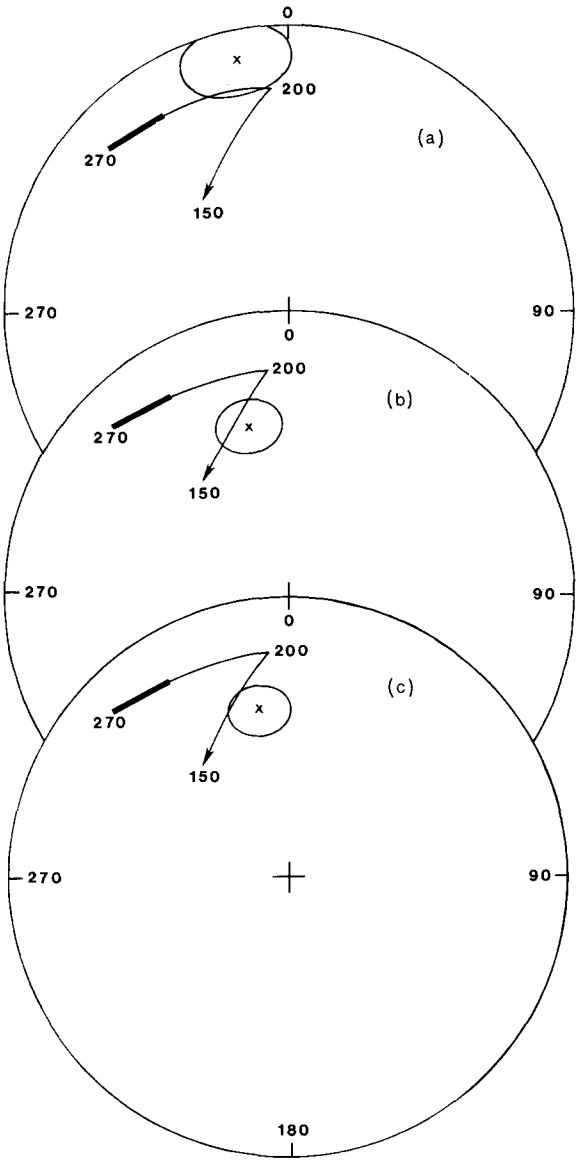


Fig. 4. Mean direction and 95% confidence circle for Koipato basalt flows, South Tobin Range, Nevada, superimposed on expected directions calculated from APW path for North America [after 14,15]. (a) In-situ mean; (b) mean after simple tilt correction (site-mean directions rotated about local strike by amount of local dip); (c) complete tilt correction (plunge of 195°/25° removed, followed by removal of residual tilt). Heavy line indicates portion of APW path corresponding to probable age of Koipato Formation.

pole-space, assuming a 95% confidence circle around the Gordon et al. [14] reference pole of 4.0°.) Note that discordance is not an artifact of the choice of reference pole; all other published 250 Ma reference poles are similar and would yield virtually identical results. The calculations above assume (1) a 250 Ma age of magnetization, and (2) that the "complete" method of tilt correction (removal of plunge followed by removal of residual tilt) is correct. However, from Fig. 4 it is apparent that the Koipato direction is discordant unless (1) it is post-folding and latest Triassic or earliest Jurassic in age (Fig. 4a), or (2) pre-folding but roughly mid-Jurassic (Fig. 4b, c). These are highly unlikely scenarios, so it clearly is worthwhile to examine possible causes for discordance.

4. Possible causes of discordance

In this section we will consider discordance from a paleomagnetic point-of-view, and we will avoid discussing any but the simplest geological ramifications. A discussion of the tectonic significance of these data will follow (R.C. Speed, manuscript in preparation).

4.1. Geomagnetic explanations

Possibly the least likely explanation for the discordance is that our mean direction has not averaged non-dipole elements of the geomagnetic field, and thus cannot be compared directly to dipole directions for the craton. We cannot estimate the time-interval represented by our 11 flows, but they span at least one polarity transition, are spread over a reasonably large geographical area, and have scatter characteristics similar to other paleomagnetic data sets that average the non-dipole field. The shape of the data set (and a consistency-of-reversal test) argue that we have not sampled an excursion or polarity transition. Thus, baring some highly improbable (and essentially untestable) abberation of the geomagnetic field, the Koipato discordance is caused by geology, not geomagnetism.

4.2. Explanations based on remagnetization

The hypothesis that the Koipato discordance is due to remagnetization is only slightly more appealing. The best arguments against remagnetization already have been made; all of the paleomagnetic evidence (previous section) suggests that the Koipato magnetization is pre-folding, and probably primary. Because remagnetization destroys any record of the paleohorizontal at

the time magnetization was acquired, the remagnetization hypothesis tends to be a Pandora's Box from which spring an infinity of explanations, most of them untestable. In this section we will examine some of the least improbable examples, with the aid of Fig. 4.

(1) If the Koipato in the southern Tobin Range was folded in situ and then remagnetized (and has not been further disturbed), Fig. 4a indicates that the rocks must have acquired their magnetization sometime before about 200 Ma; Late Triassic or Early Jurassic are the best bets. However, there are no known events of that age (e.g., intrusions, regional metamorphism, mineralization) that can account for remagnetization—and a positive fold test indicates that the magnetization is pre-folding.

(2) Possibly the Koipato was tilted as it formed and was remagnetized shortly thereafter. That is, perhaps the flows were extruded, solidified, and tilted in rapid succession, and then magnetization was "frozen in" as a result of regional cooling. In this case the rocks should retain a direction similar to the heavy portion of the APW path of Fig. 4, and a few degrees of clockwise rotation would give the direction shown in Fig. 4a. This explanation suffers from all the deficiencies of (1) except the need to identify a cause of remagnetization. It also requires a later event to produce the rotation.

(3) If the rocks were remagnetized before folding and while still horizontal, then Fig. 4b and c show that this must have happened sometime about 175 Ma (Middle Jurassic) if the Koipato rocks have not moved relative to North America. But again there are no known events of this age to account for remagnetization, and all the magnetic evidence argues against it.

(4) Widespread intrusion during the Middle or Late Cretaceous led to considerable remagnetization in the Sierra Nevada region and in western Nevada. Because remagnetization obscures the paleohorizontal, the following-scenario is possible (although in our view, very unlikely): (a) formation of the Koipato in the Late Permian or Early Triassic; (b) deformation at some time thereafter, but pre-Late Cretaceous; (c) complete remagnetization in the Late Cretaceous (Middle Cretaceous is not possible because of the reversed-polarity flows), at a time when the regional structure was a homocline dipping about 30° to the north; (d) no tectonic disturbance thereafter. But here again the magnetic evidence argues against remagnetization —and the level of coincidence needed to make this explanation work is very high.

4.3. Structural explanations

The Koipato magnetic discordance consists mainly of a roughly 20° steepening of inclination (95% error limits about 10°). Thus the discordance would be nicely accounted for if the rocks had accumulated on a regional southward slope of about 20°, instead of a horizontal surface. This is the troublesome "original dip" problem of paleomagnetism, which in many cases can be overcome simply by sampling over an area large enough to average out paleotopographic irregularities. Unfortunately, we have data for only a fairly restricted part of a single mountain range. An examination of situations where basalt flows occur suggests that original dips of 20° are perhaps a bit extreme, but that in worst-case scenarios 15° is possible. For instance, south slopes of modern Cascade volcanos reach 16–17° but average somewhat less. Shield volcanos have much lower slopes (e.g., the 1929 flow on the south flank of Mauna Loa flowed down a slope of slightly less than 7°). The maximum regional slope down east-facing scarps in the Death Valley area is about 15°, but the average is much less. Maximum slopes into most sedimentary basins are very small.

In view of this, it appears that most of the inclination-discordance seen in the southern Tobin Range Koipato data could reflect original dip, but probably does not. Most basalts accumulate on slopes of a few degrees or less, even basalts forming as parts of calc-alkaline magmatic arcs (where steep-sided volcanos are the most conspicuous topographic features). Large regional slopes imply great topographic relief, and high-relief features usually are eroded away; rarely, we hope, are they preserved in the geologic record to deceive honest paleomagnetists! Thus we suspect that the Koipato discordance cannot be attributed to original dip, but we cannot be absolutely sure. More Koipato data, spread over a larger area, are required.

4.4. Explanations based on regional tectonics

If the Koipato magnetization is primary and was acquired on a nearly horizontal surface, and if none of the "geomagnetic" alternatives are tena-

ble, then the southern Tobin Range was transported to Western Nevada from far to the north. That is, after magnetization in Permo-Triassic time, a crustal fragment including the Koipato flows moved southward, relative to North America, a long distance. How far it moved is not very well constrained by paleomagnetism; the most probable distance, from the *P*-values given earlier, is about 1400 km. On the way, or subsequently, the crustal fragment also underwent a net clockwise rotation of about 25°. (Minimum values, taking the 95% confidence limits at face value, are about 650 km and 18°, respectively.) Rotation could have happened during transport, during docking with North America, or later.

Similarly, paleomagnetism provides little control on the timing of southward transport. On the basis of paleomagnetic information it could have occurred at any time after the Koipato rocks formed; an upper limit comes from concordant Oligocene ashflow tuffs in the Great Basin [17]. Our work on rocks younger than the Koipato eventually may constrain the timing, but so far the results are inconclusive. The stratigraphic position of the Koipato volcanics in the southern Tobin Range suggests that the Golconda allochthon, and possibly the post-Koipato Triassic cover strata, moved with the Koipato. Thus the time of transport may have been post-Triassic, supporting the idea of a mid-Mesozoic attachment of at least part of the Golconda allochthon [5].

5. Conclusions

Koipato basalts of the southern Tobin Range, Nevada, retain a remanent magnetization that almost certainly is pre-folding, and probably primary. The mean direction of magnetization of these rocks is strongly discordant with respect to the APW path of North America. The most likely explanation for this discordance is relative southward transport and clockwise rotation. Judging from the paleomagnetic evidence, the Koipato rocks) and the Golconda rocks upon which they rest) probably are allochthonous and far-travelled, and very possible exotic to North America. The sense and amount of motion called for are very similar to values given by Maher and Jirschvink [18] for mid-Jurassic rocks in the Jackson Mountains of northwest Nevada. On a broader level, southward displacement of the South Tobin Range is excellent confirmation of the concept of relative southward migration of Cordilleran "transpressional terranes" during the early Mesozoic, as proposed by Ave Lallemant and Oldow [20]. The mechanism and timing of this displacement are discussed by Beck [21–23]. Our data suggest that the Sonoma orogeny, at least as represented in the southern Tobin Range, involved far-travelled rocks that may not have reached their present location until mid-Mesozoic time.

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